

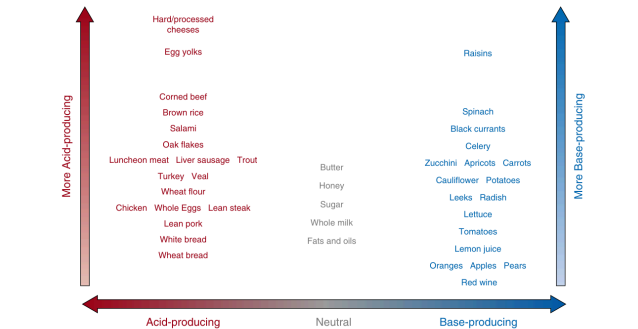


Fig. 55.1 Graphic presentation of selected common foods that are generally acid producing, neutral, or base producing (horizontal orientation) and their comparable acid- or base-producing capacity (vertical orientation). The listed neutral foods are comparable with respect to their contribution to net endogenous acid production (NEAP), so their vertical orientation does not indicate ranking with respect to NEAP.

Fig_55_1

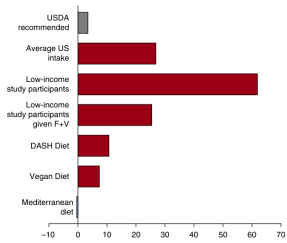


Fig. 55.2 Graphic presentation of the potential renal acid load, or net acid-producing capacity, of the indicated diets, calculated from their standard food components. The net acid-producing capacity of components of the U.S. Department of Agriculture (USDA) recommended diet is used as the standard to which to compare the net acid-producing capacity of the remaining indicated diets; it is represented by the gray bar at the top. Diets with net acid-producing capacity greater than the USDA diet are represented by red bars, and the one with less is represented by a blue bar. DASH, Dietary Approaches to Stop Hypertension; F + V, fruits and vegetables. (Data for diets of low-income study participants are from references 102, 103, and 171.)

Fig_55_2

Table 55.1 Dietary Approaches to Reduce Incidence of Type 2 Diabetes	
Strategy	Tactic
Emphasize carbohydrates that promote healthy glucose metabolism	- Promote foods with dietary fiber - Whole grains - Fruits and vegetables - Minimize foods with high glycemic index - Simple sugars - Fructose
Emphasize fats that promote healthy fat metabolism	Promote polyunsaturated fats, including omega 6's and omega 3's
Emphasize dietary protein that is less likely to increase insulin resistance	- Promote plant-based foods - Minimize animal-based foods, particularly red meat
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt") - Promote a healthy caloric intake - Maximize foods with high-nutrition/low-calorie ratio like fruits and vegetables - Minimize processed foods
Reduce obesity risk	Promote omega-3 polyunsaturated fats

Table_55_1

Table 55.2 Dietary Approaches to Reduce Incidence of Hypertension	
Strategy	Tactic
Emphasize a "healthy" profile of electrolyte intake	- Minimize NaCl ("salt") - Fresh rather than processed foods - Fresh fruits and vegetables - Promote potassium intake as tolerated - Fruits and vegetables
Emphasize proteins that promote healthy blood pressure	- Promote magnesium intake as tolerated - Fruits and vegetables - Promote plant-based foods - Minimize animal-based foods
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Reduce obesity risk	- Promote a healthy caloric intake - Maximize foods with high nutrition/low-calorie ratio like fruits and vegetables - Minimize processed foods

Table_55_2

Table 55.3 Dietary Approaches to Reduce the Risk for Initiation of Chronic Kidney Disease in Susceptible Individuals	
Strategy	Tactic
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables - Minimize animal-based foods - Minimize NaCl ("salt")
Emphasize an overall diet with a high ratio of fiber-to-animal-based protein	- Promote plant-based foods - Minimize animal-based foods, particularly red meat
Emphasize fats that enhance kidney health	Promote polyunsaturated fats including omega 3's

Table_55_3

Table 55.4 Dietary Approaches to Reduce the Risk for Progression of Established Chronic Kidney Disease (CKD)	
Strategy	Tactic
Emphasize an overall diet that minimizes net endogenous acid production	- Promote base-producing fruits and vegetables as tolerated - Minimize animal-based foods - Minimize NaCl ("salt") - Minimize NaCl ("salt") intake
Emphasize electrolyte intake that lowers risk for CKD progression	
Emphasize an overall diet that yields a gut microbiome which promotes kidney health	- Promote plant-based foods as tolerated - Minimize animal-based foods, particularly red meat

Table_55_4

Table 55.5 Dietary Approaches to Reduce Mortality in Established Chronic Kidney Disease	
Strategy	Tactic
Emphasize an overall diet with a high ratio of fiber-to-animal-based protein	- Promote plant-based foods - Minimize animal-based foods, particularly red meat - Maintain protein intake that avoids protein energy wasting
Minimize dietary NaCl ("salt")	- Promote fresh fruits and vegetables as tolerated - Minimize processed foods - Minimize NaCl ("salt") intake
Emphasize diets that improve blood pressure control	- Emphasize potassium intake as tolerated - Emphasize diets that reduce net endogenous acid production

Table_55_5

Table 55.6 General Approach for a Dietary Recommendation for an Individual with Chronic Kidney Disease and Reduced Glomerular Filtration Rate	
Dietary Consideration	Recommendation
Energy	30-35 kcal/kg body wt/day 5%-15% from carbohydrates - Avoid fructose
Protein	0.7-0.8 g ideal body wt/day for nondialysis patients 1.2 g ideal body wt/day for dialysis patients - Nontitrated ketonurics might be substituted for a portion of recommended dietary protein - Plant-based protein can be considered for patients able to tolerate the additional potassium load
Sodium	<2 g 90 mmol/day unless otherwise contraindicated
Potassium	~4 g (102 mmol/day) for CKD stages 1 and 2 2-4 g (51-102 mmol/day) for CKD stages 3 and 4 - As instructed by provider for CKD stage 5
Phosphorus	~800 mg/day - Promote plant-based sources over animal-based sources
Calcium	- Minimize processed foods 1000 mg 25 mmol/day for adults aged 18-50 years 1200 mg 30 mmol/day for adult females aged 51-70 years 1200 mg 30 mmol/day for adults >70 years
Fiber	>25 g/day
Micro nutrients	- Promote a primarily plant-based diet - Multivitamin supplement for dialysis patients

Table_55_6

Table 55.7 Dietary Interventions That Help in Management of Chronic Kidney Disease-Related Complications	
Strategy	Tactic
Nephropathy progression	- Blood pressure control - Glycemic control in patients with diabetes - Reduce net endogenous acid production - Promote a "kidney-friendly" gut microbiome
Hypertension	- Minimize NaCl ("salt") - Promote plant-based food as tolerated - Potassium as tolerated - Magnesium as tolerated
Disturbed calcium/phosphate metabolism	Promote plant-based food as tolerated
Metabolic acidosis	- Minimize processed foods - Promote base-producing fruits and vegetables as tolerated - Minimize animal-based protein - Minimize NaCl ("salt")
Increased protein catabolism	Treat metabolic acidosis with the described dietary approaches

Table_55_7

Table 55.8 Common Components of Various Diets That Might Be Prescribed				
	Mediterranean Diet	Mediterranean Kidney Diet	DASH Diet	Plant-Based Low-Protein (PLADO) Diet
Cereals/whole grains	Every meal	Every meal	7-8 servings/day	Every meal
Fruits + veggies	3-4 servings/day	K friendly F+V*	4-5 servings/day	3-5 servings/day
Olive oil	Every meal	Every meal	2-3 servings/day	2-3 servings/day
Fish/Seafood	5-6 times/week	5-6 times/week	0-2 servings/day	Minimal 14-32 g/day
Meat—white	0-2 times/week	0-2 times/week	0-2 servings/day	Minimal
Meat—Red, lean	<4 times/month	<4 times/month	0-2 servings/day	Minimal
Nuts/seeds	1-2 times/day	1-2 times/day*	1-2 servings/day	3-4 servings/day
Sweets	<3/week	<3/week	<5/week	3-5/week
Dairy	2 servings/day	2 servings/day	2-3 servings/day	2-3 servings/day

Table_55_8